Hypervalinemia and Hyperleucine-Isoleucinemia (MONDO:0100058): A Comprehensive Disease Characterization

Disease category: Genetic (inborn error of metabolism) **Primary gene:** *BCAT2* (mitochondrial branched-chain aminotransferase, BCATm) **Inheritance:** Autosomal recessive **MONDO ID:** MONDO:0100058

Summary

Hypervalinemia and Hyperleucine-Isoleucinemia (MONDO:0100058) is an ultra-rare autosomal recessive inborn error of branched-chain amino-acid (BCAA) metabolism caused by biallelic loss-of-function mutations in the *BCAT2* gene, which encodes the mitochondrial branched-chain aminotransferase (BCATm; EC 2.6.1.42). This enzyme catalyzes the first, shared catabolic step for all three BCAAs — valine, leucine and isoleucine — in which each amino acid is reversibly transaminated with α-ketoglutarate to yield the corresponding branched-chain keto acid (BCKA) plus glutamate. Because the enzymatic block lies **upstream** of the branched-chain α-keto-acid dehydrogenase (BCKDH) complex that is deficient in maple syrup urine disease (MSUD), BCAT2 deficiency elevates plasma valine, leucine and isoleucine **without overproducing the neurotoxic branched-chain keto acids**, yielding a comparatively mild, chronic neurometabolic phenotype rather than the fulminant ketoacidotic crises of MSUD.

The disease nosology unifies two historically distinct aminoacidopathies first described in the 1960s — classic Hypervalinemia (OMIM 277100; Wada et al. 1963) and Hyperleucine-isoleucinemia — under a single modern molecular entity. The molecular basis was established only in 2015, when Wang et al. (PMID: 25653144) reported the first and to date only molecularly confirmed patient: a 25-year-old man with compound heterozygous *BCAT2* variants c.509G>A (p.Arg170Gln, paternal) and c.790G>A (p.Glu264Lys, maternal), both shown by functional assay to reduce BCAT2 enzyme activity. The patient presented with ~6 years of headache and mild memory impairment, symmetric cerebral white-matter abnormalities on MRI, and markedly elevated plasma valine and leucine. Notably, **treatment with vitamin B6**

(pyridoxine) — the cofactor precursor of the enzyme's pyridoxal-5′-phosphate prosthetic group — lowered BCAAs and improved the MRI lesions, establishing the disorder as B6-responsive and dietarily manageable.

A *Bcat2* (BCATm) knockout mouse independently validates the mechanism: these animals exhibit the biochemical hallmark of elevated plasma BCAAs, but the dominant phenotype is metabolic (leanness, increased energy expenditure, improved glucose tolerance via a futile protein-turnover cycle) rather than neurological. Both human causal variants are ultra-rare in gnomAD (MAF ~1×10^{-5} or lower, no homozygotes), consistent with an ultra-rare recessive disorder. This report presents the complete causal chain, genetic and biochemical details, phenotype and HPO mapping, diagnostic strategy, treatment, prognosis, prevention, model organisms, and the substantial knowledge gaps that remain given that human evidence rests on a single case.

Key Findings

Finding 1 — Biallelic *BCAT2* mutations are the molecular cause (autosomal recessive)

The defining genetic lesion is loss of function of *BCAT2*. Wang et al. 2015 (PMID: 25653144) reported the first human case — a 25-year-old man carrying two *BCAT2* variants in compound-heterozygous configuration: **c.509G>A (p.Arg170Gln)** inherited from the father and **c.790G>A (p.Glu264Lys)** inherited from the mother. The abstract states directly that "*two heterogeneous BCAT2 gene mutations were found in the patient, including c.509G > A (p.Arg170Gln) and c.790G > A (p.Glu264Lys),*" and that these "*were found in his father and mother, respectively, suggesting an autosomal recessive disorder*." Critically, the pathogenicity was functionally established: "*BCAT2 functional studies demonstrated that the two BCAT2 gene mutations resulted in decreased BCAT2 enzyme activity.*" This provides both the segregation evidence for autosomal recessive inheritance and the functional (PS3-grade) evidence that the variants are loss-of-function.

Finding 2 — BCAT catalyzes the first, shared step of BCAA catabolism; its loss elevates plasma BCAAs and injures the CNS

Valine, leucine and isoleucine are essential BCAAs whose catabolism begins with a single transamination step catalyzed by branched-chain aminotransferase (BCAT). Wang et al. establish the pathophysiologic principle: "*When BCAA metabolism is genetically impaired in human, serum levels of BCAA and/or their metabolites rise considerably, causing severe neurological dysfunction. The first step in BCAA catabolism is catalyzed by branched chain aminotransferase (BCAT).*" In the index patient this manifested biochemically and radiologically: "*Brain MRI showed symmetric white matter abnormal signals. Metabolic studies revealed remarkably elevated plasma valine and leucine concentrations.*" The combination localizes the disease to the central nervous system — specifically bilateral symmetric cerebral white matter — against a backdrop of systemic BCAA accumulation.

Finding 3 — Vitamin B6 (pyridoxine) is therapeutically effective

BCAT2 is a pyridoxal-5′-phosphate (PLP)–dependent enzyme, and the disorder is cofactor-responsive. After pyridoxine supplementation, "*the levels of BCAA, especially valine were remarkably decreased and brain MRI lesions were improved*" (PMID: 25653144). This dual biochemical-and-radiological response is clinically important: it indicates that residual enzyme function can be pharmacologically augmented by saturating the apoenzyme with its PLP cofactor, and that the white-matter pathology is at least partially reversible when BCAA levels are normalized.

Finding 4 — The *Bcat2* knockout mouse recapitulates the biochemistry but not the neurology

She et al. 2007 (PMID: 17767905) generated a global *BCATm* (*Bcat2*) knockout and confirmed enzyme identity: "*we disrupted the BCATm gene, which encodes the enzyme catalyzing the first step in peripheral BCAA metabolism.*" The knockouts reproduce the biochemical hallmark but present a predominantly metabolic phenotype: "*BCATm(-/-) mice exhibited elevated plasma BCAAs and decreased adiposity and body weight, despite eating more food, along with increased energy expenditure, remarkable improvements in glucose and insulin tolerance, and protection from diet-induced obesity.*" The mechanism is a futile protein-turnover cycle. Lynch et al. 2015 (PMID: 26351290) independently confirmed the model, noting that "*Mice lacking the mitochondrial branched-chain aminotransferase (BCATm/Bcat2), which interconverts leucine and*

α-ketoisocaproate, exhibit elevated protein turnover," with transcriptomic upregulation of mTOR/eIF/protein-synthesis and degradation pathways in skeletal muscle. The divergence from the human neurological phenotype is itself informative (see Limitations).

Finding 5 — Disease nosology bridges 1960s biochemistry and modern genetics

MONDO:0100058 unifies two classic aminoacidopathies. **Classic Hypervalinemia** (OMIM 277100) was first described by Wada et al. in 1963 (*"Idiopathic hypervalinemia: probably a new entity of inborn error of valine metabolism"*; PMID: 14077060) and attributed to deficient valine transamination. **Hyperleucine-isoleucinemia** was the corresponding leucine/isoleucine transamination defect. Wang et al. 2015 (PMID: 25653144) provided the molecular unification, showing that biallelic *BCAT2* mutations produce combined elevation of valine, leucine and isoleucine — exactly what would be expected if a single aminotransferase acting on all three substrates is deficient.

Finding 6 — Mechanistic distinction from MSUD explains the milder phenotype

BCAT catalyzes the first catabolic step (Val/Leu/Ile + α-ketoglutarate ⇌ BCKAs [α-ketoisovalerate/KIV, α-ketoisocaproate/KIC, α-keto-β-methylvalerate/KMV] + glutamate). The **downstream** second step is the BCKDH complex, deficiency of which causes MSUD. In MSUD, both BCAAs **and** the neurotoxic BCKAs accumulate, driving recurrent metabolic crises, cerebral edema, and neurodevelopmental impairment: "*Maple Syrup Urine Disease (MSUD) is a rare autosomal recessive metabolic disorder characterised by defective branched-chain amino acid (BCAA) catabolism, leading to neurotoxicity, recurrent metabolic crises, and neurodevelopmental impairment*" (PMID: 42329506), with leucine identified as "*the primary neurotoxic metabolite*" (PMID: 38682838). In BCAT2 deficiency the block is **upstream** of BCKDH, so BCAAs accumulate but BCKAs are *not* overproduced. This explains why the single reported patient had an indolent adult-onset leukoencephalopathy rather than neonatal ketoacidotic crises, and why MSUD was genetically excluded.

Finding 7 — Both causal variants are ultra-rare in gnomAD

gnomAD v4 (queried 2026-10) confirms both variants are ultra-rare with no reported homozygotes, supporting classification as pathogenic in an ultra-rare recessive disorder:

Variant (protein)	cDNA	genomic (GRCh38)	gnomAD exome MAF	gnomAD genome
p.Glu264Lys	c.790G>A	19-48797239-C-T	~1.2×10^{-5} (18/1,461,568)	2/152,310
p.Arg170Gln	c.509G>A	19-48800003-C-T	~1.4×10^{-6} (2/1,461,592)	absent

The population rarity (PM2) combined with the in vitro functional evidence (PS3: "*the two BCAT2 gene mutations resulted in decreased BCAT2 enzyme activity*," PMID: 25653144) supports a pathogenic classification under ACMG/AMP criteria.

Finding 8 — Diagnostics, treatment, prognosis and prevention

Diagnosis rests on **quantitative plasma amino-acid analysis** showing elevated valine, leucine and isoleucine ("*remarkably elevated plasma valine and leucine concentrations*"), **brain MRI** showing symmetric white-matter signal abnormalities, **exclusion of MSUD** ("*Maple syrup urine disease (MSUD) diagnosis was not supported since all genes for the branched-chain α-keto acid dehydrogenase complex (BCKD) gene were normal*"), and **molecular confirmation** of biallelic *BCAT2* variants by WES/WGS or a metabolic gene panel, optionally supported by an in vitro BCAT2 enzyme-activity assay (PMID: 25653144). Treatment is **vitamin B6 (pyridoxine)**, which normalized BCAAs and improved MRI lesions; **dietary BCAA/protein restriction** is a rational adjunct by analogy to MSUD management (PMID: 42329506). A PubMed search found no further molecularly confirmed *BCAT2* cases since 2015.

Finding 9 — Anatomical, cellular and subcellular localization

The pathology localizes to **bilateral symmetric cerebral white matter** (UBERON:0002316): "*Brain MRI showed symmetric white matter abnormal signals*" (PMID: 25653144), with neurocognitive symptoms (chronic headache, mild memory impairment). **Subcellularly, BCAT2/BCATm is a mitochondrial-matrix enzyme** (She 2007: "*mitochondrial branched-chain aminotransferase*"; PMID: 17767905), ubiquitously expressed with high activity in peripheral tissues. The brain additionally relies on BCAT (predominantly the cytosolic BCAT1 isoform) for glutamate-nitrogen shuttling, making the CNS particularly sensitive to BCAA dyshomeostasis.

Mechanistic Model / Causal Chain

The pathophysiology proceeds as an ordered causal chain from the germline mutation to the clinical manifestation. Steps marked *(inferred)* are reasoned by analogy or from the mouse model rather than directly demonstrated in the single human patient.

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1. Biallelic loss-of-function BCAT2 variants (p.Arg170Gln / p.Glu264Lys)
        │  leads to
        ▼
2. Reduced mitochondrial branched-chain aminotransferase (BCATm) enzyme
   activity  [demonstrated: in vitro activity assay, <a href="https://pubmed.ncbi.nlm.nih.gov/25
        │  results in
        ▼
3. Impaired FIRST catabolic step of BCAA metabolism:
   Val / Leu / Ile + α-ketoglutarate  ⇌  BCKA + glutamate  (reaction blocked)
        │  results in
        ▼
4. Accumulation of plasma valine, leucine and isoleucine
   (BCKAs are NOT overproduced — block is upstream of BCKDH)   ◄── KEY BRANCH-POINT
   │                                                          │
   │ (contrast: MSUD = downstream BCKDH block →               │
   │  BOTH BCAA *and* neurotoxic BCKA accumulate →            │
   │  crises, cerebral edema, <a href="https://pubmed.ncbi.nlm.nih.gov/42329506/" rel="noopener
        │  leads to
        ▼
5. Disturbed cerebral BCAA/glutamate nitrogen homeostasis (inferred:
   brain depends on BCAT for glutamate-nitrogen shuttling)
        │  results in
        ▼
6. Symmetric cerebral white-matter injury (leukoencephalopathy)
   [demonstrated: MRI symmetric white-matter signal, <a href="https://pubmed.ncbi.nlm.nih.gov/25
        │  manifests as
        ▼
7. Chronic, indolent neurological phenotype: headache, mild memory impairment
   (adult-onset; NOT neonatal ketoacidotic crisis)

   ── THERAPEUTIC REVERSAL ──
   Vitamin B6 (pyridoxine) → ↑ PLP cofactor saturation of residual BCATm →
   ↓ plasma BCAAs (esp. valine) → MRI lesion improvement [PMID 25653144]
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Upstream vs downstream, and the MSUD comparison

Feature	BCAT2 deficiency (this disease)	MSUD (differential)
Defective enzyme	BCAT2 / BCATm (transaminase)	BCKDH complex (dehydrogenase)
Pathway position	First step (upstream)	Second step (downstream)
BCAAs	Elevated (Val, Leu, Ile)	Elevated
Branched-chain keto acids (BCKAs)	**Not overproduced**	**Accumulate (neurotoxic)**
Alloisoleucine	Absent	Present (pathognomonic)
Onset / course	Adult, chronic, indolent	Neonatal, acute crises
Hallmark injury	Symmetric leukoencephalopathy	Cerebral edema, encephalopathy
Cofactor	PLP (vitamin B6)	Thiamine (for thiamine-responsive forms)
Treatment	Vitamin B6 + diet	Strict protein restriction, BCAA-free formula

Pathway and ontology annotations. Biochemical pathway: branched-chain amino acid degradation (KEGG hsa00280; Reactome "Branched-chain amino acid catabolism"). Enzyme: EC 2.6.1.42, PLP-dependent. Gene Ontology biological process: GO:0009083 (branched-chain amino acid catabolic process), GO:0009081 (branched-chain amino acid metabolic process), GO:0009448 (gamma-aminobutyric acid metabolic process — nitrogen shuttling context). GO cellular component: GO:0005759 (mitochondrial matrix). Chemical entities (CHEBI): L-valine (CHEBI:16414), L-leucine (CHEBI:15603), L-isoleucine (CHEBI:17191), pyridoxal 5′-phosphate (CHEBI:18405), pyridoxine/vitamin B6 (CHEBI:16709), α-ketoglutarate/2-oxoglutarate (CHEBI:16810), L-glutamate (CHEBI:16015). Cell types of interest (CL): oligodendrocyte (CL:0000128) and its myelin, astrocyte (CL:0000127) for CNS glutamate handling.

Disease Characterization by Template Section

1. Disease Information

A concise overview: an ultra-rare autosomal recessive inborn error of BCAA metabolism in which deficiency of mitochondrial branched-chain aminotransferase (BCAT2/BCATm) blocks the first catabolic step shared by valine, leucine and isoleucine, elevating all three in plasma and producing a chronic leukoencephalopathy. **Identifiers:** MONDO:0100058; classic Hypervalinemia OMIM 277100; gene *BCAT2* OMIM *113530; HGNC:977; NCBI Gene 587; UniProt O15382; chromosome 19q13.33. MeSH: "Hypervalinemia" (within amino acid metabolism, inborn errors). Synonyms: Hypervalinemia; Hyperleucine-isoleucinemia; branched-chain aminotransferase (BCAT2) deficiency; BCATm deficiency; valine transaminase deficiency (historical). Information source:** aggregated disease-level resources (OMIM, MONDO) plus a single published patient case (not EHR-derived).

2. Etiology

Causal factor: genetic — biallelic loss-of-function *BCAT2* variants. **Genetic risk factors:** the only known causal variants are p.Arg170Gln (c.509G>A) and p.Glu264Lys (c.790G>A); by analogy, consanguinity/carrier status in families would be the relevant risk context. **Environmental risk/modifying factors:** dietary BCAA (protein) intake plausibly modulates biochemical severity (inferred from MSUD and the mouse model). **Protective factors:** vitamin B6 status is protective by augmenting residual enzyme; low-BCAA diet is protective by reducing substrate load. **Gene-environment interaction:** the disorder is cofactor- and diet-responsive, a classic gene-nutrient interaction (genotype determines residual enzyme; B6 and dietary BCAA load determine expressed phenotype).

3. Phenotypes

From the single human case (PMID: 25653144):

Phenotype	Type	HPO suggestion	Onset/ course	Frequency
Elevated plasma valine (hypervalinemia)	Lab abnormality	HP:0500152 (hypervalinemia) / HP:0003355 (aminoaciduria/amino acid metab abnormality)	Present at diagnosis	1/1 case
Elevated plasma leucine/ isoleucine	Lab abnormality	HP:0010903 (abnormal circulating amino acid concentration)	Present	1/1
Headache	Symptom	HP:0002315	~6 yr chronic	1/1
Memory impairment (mild)	Cognitive sign	HP:0002354	Chronic	1/1
Cerebral white-matter abnormality (symmetric leukoencephalopathy)	Imaging/ physical	HP:0002500 (abnormality of cerebral white matter) / HP:0007024	Chronic, reversible with B6	1/1

Severity: mild-to-moderate; progression: chronic/indolent, partially reversible with treatment. **Quality-of-life impact:** modest in the reported case (chronic headache, mild memory complaints) — far milder than MSUD, which carries high morbidity/mortality. Note: frequencies are drawn from a single patient and cannot be generalized.

4. Genetic / Molecular Information

Causal gene: *BCAT2* (OMIM *113530; HGNC:977; UniProt O15382; chr19q13.33). Pathogenic variants: c.509G>A (p.Arg170Gln) and c.790G>A (p.Glu264Lys), both missense, both reducing enzyme activity (loss-of-function). Classification: pathogenic/likely pathogenic by PS3 (functional) + PM2 (ultra-rare, gnomAD) + segregation. Allele frequency: ultra-rare (MAF ≤~1.2×10⁻⁵; no homozygotes; see Finding 7 table). Origin: germline. Functional consequence: loss of function (reduced transaminase activity). Modifier genes: none established;* BCAT1 *(cytosolic isoform) is a plausible functional modifier via compensation but is unproven. Epigenetic / chromosomal abnormalities:** none reported or expected for this point-mutation recessive disorder.

5. Environmental Information

Dietary BCAA/protein is the principal non-genetic modulator of biochemical severity. **Vitamin B6 (pyridoxine)** status influences residual enzyme activity. No toxins, radiation, occupational exposures, or infectious agents are implicated. (The index patient's leukoencephalopathy was metabolic, not infectious or toxic.)

6. Mechanism / Pathophysiology

See the full causal chain and MSUD-comparison table above. Key points: (a) molecular pathway = BCAA catabolism (KEGG hsa00280); (b) biochemical defect = PLP-dependent transaminase (EC 2.6.1.42) loss of function; (c) subcellular compartment = mitochondrial matrix (GO:0005759); (d) metabolic change = combined Val/Leu/Ile accumulation *without* BCKA overproduction; (e) tissue damage = cerebral white-matter injury, partially reversible. The mouse model adds a peripheral metabolic dimension (futile protein-turnover cycle, increased energy expenditure; PMID: 17767905, PMID: 26351290).

7. Anatomical Structures Affected

Primary organ: brain — bilateral symmetric cerebral white matter (UBERON:0002316). **Body system:** central nervous system (UBERON:0001017). **Cell/tissue level:** oligodendrocytes/myelin (CL:0000128) implicated by the white-matter signal; astrocytes (CL:0000127) relevant for CNS glutamate-nitrogen handling. **Subcellular:** mitochondrial matrix (GO:0005759), the locus of BCATm. **Lateralization:** bilateral/symmetric. Peripherally, skeletal muscle and adipose tissue are major sites of BCATm activity (from mouse data) but human peripheral pathology is not documented.

8. Temporal Development

Onset: adult (the index case presented at ~25 years after ~6 years of symptoms); classic infantile hypervalinemia, historically, had pediatric presentation. **Onset pattern:** insidious/chronic. **Progression:** slow, non-crisis; partially reversible with vitamin B6. **Disease course:** chronic, lifelong (genetic), but manageable. No evidence of relapsing-remitting crises (contrast MSUD). **Critical period for intervention:** B6 therapy and dietary management appear effective once diagnosed; earlier intervention is presumed better by analogy.

9. Inheritance and Population

Inheritance: autosomal recessive (parental carrier segregation, PMID: 25653144). **Prevalence/incidence:** unknown; ultra-rare — only one molecularly confirmed patient worldwide plus a handful of historical biochemical cases. **Penetrance/expressivity:** cannot be estimated from n=1. **Carrier frequency:** both variants are ultra-rare in gnomAD, implying very low carrier frequency. **Founder effects/consanguinity:** none documented. **Sex ratio / geographic distribution / age distribution:** undetermined given the single case (reported from China).

10. Diagnostics

Laboratory: quantitative plasma amino acids (elevated Val, Leu, Ile). **Imaging:** brain MRI (symmetric white-matter signal abnormality). **Confirmatory:** molecular testing — WES/WGS or inborn-errors-of-metabolism gene panel including *BCAT2*; single-gene *BCAT2* sequencing; optional in vitro BCAT2 enzyme-activity assay. **Differential diagnosis — MSUD:** exclude by demonstrating normal BCKDHA/BCKDHB/DBT genes and absence of alloisoleucine and elevated BCKAs ("*all genes for the branched-chain α-keto acid dehydrogenase complex (BCKD) gene were normal*," PMID: 25653144). **Screening:** elevated BCAAs are detectable by newborn-screening tandem mass spectrometry (same analyte panel used for MSUD), making biochemical screening feasible, though the disorder's rarity means it is not specifically targeted.

11. Outcome / Prognosis

Prognosis in the single documented case was favorable: biochemical normalization and MRI improvement on vitamin B6. In sharp contrast to MSUD — where decompensation causes cerebral edema and death (PMID: 41257986, PMID: 38682838) — BCAT2 deficiency lacks the neurotoxic BCKA accumulation and is not associated with acute metabolic crises. Mortality, survival and quality-of-life data at the population level are unavailable (n=1). Prognostic factors are presumed to include residual enzyme activity, B6-responsiveness, and dietary adherence.

12. Treatment

Pharmacotherapy / cofactor: vitamin B6 (pyridoxine) — demonstrated to lower BCAAs and improve MRI lesions (PMID: 25653144); NCIT suggestion: Pyridoxine (C738)/Vitamin B6 Therapy. **Dietary:** BCAA/protein restriction, by analogy to MSUD management (PMID: 42329506); NCIT: Dietary Intervention (C15222). **Supportive:** symptom management (headache). **Experimental / gene/cell/RNA therapy:** none reported for this disorder

specifically. (The MSUD oral leucine-decarboxylase enzyme-therapy work, PMID: 37494004, is a conceptual neighbor but not applicable here, since BCAT2 deficiency does not overproduce leucine metabolites.) **Personalized medicine:** genotype- and B6-response-guided management.

13. Prevention

Primary: genetic counseling for carrier couples; the disorder is single-gene recessive, so cascade carrier testing and reproductive options (prenatal/preimplantation genetic testing) apply. **Secondary:** biochemical newborn screening (tandem MS already detects elevated BCAAs) enables early detection; dietary and B6 intervention can then forestall white-matter injury. **Tertiary:** ongoing B6 + dietary management to prevent progression of leukoencephalopathy. No immunization or public-health/vector interventions apply.

14. Other Species / Natural Disease

Taxonomy / orthologs: human *BCAT2* (NCBI Gene 587) has a direct ortholog in mouse *Bcat2* (NCBI Gene 12036) and across mammals; the enzyme and pathway are deeply conserved. **Natural disease in other species:** no naturally occurring *BCAT2*-deficiency disease is documented in companion animals or wildlife (no OMIA entry identified). **Comparative biology:** the mouse knockout shows a metabolic (not neurological) phenotype, highlighting species differences in CNS BCAA handling. **Zoonotic potential:** not applicable (genetic disease).

15. Model Organisms

Mouse (*Mus musculus*, NCBI Taxon 10090): global *Bcat2*/*BCATm* knockout (She et al. 2007, PMID: 17767905; Lynch et al. 2015, PMID: 26351290). **Model type:** mammalian, genetic knockout. **Phenotype recapitulation:** reproduces the biochemical hallmark (elevated plasma BCAAs) and confirms enzyme identity and pathway position, but does **not** reproduce the human neurological/leukoencephalopathy phenotype — instead showing leanness, increased energy expenditure, improved glucose/insulin tolerance, and protection from diet-induced obesity via a futile protein-turnover cycle. **Model limitations:** fails to capture CNS white-matter pathology; dominated by peripheral metabolic effects. **Applications:** studying systemic BCAA/energy metabolism, protein turnover, and the metabolic consequences of transaminase loss. **Resources:** MGI.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role
25653144	*Hypervalinemia and hyperleucine-isoleucinemia caused by mutations in BCAT2*	Human clinical (n=1)	**Defining paper** — causal variants, AR inheritance, functional LOF, MRI phenotype, B6 response, MSUD exclusion
14077060	*Idiopathic hypervalinemia: probably a new entity* (Wada 1963)	Human clinical (historical)	Original description of classic Hypervalinemia (OMIM 277100)
17767905	*Disruption of BCATm in mice → futile protein turnover*	Model organism	Mouse KO: biochemical recapitulation, metabolic phenotype, enzyme identity
26351290	*Global deletion of BCATm increases protein-turnover genes*	Model organism	Independent confirmation of KO model and enzyme function
42329506	*Treatment/outcomes in paediatric MSUD*	Human clinical (differential)	Defines downstream MSUD; supports dietary-restriction rationale
41257986	*Cerebral edema in MSUD*	Human clinical (differential)	Illustrates BCKA-driven neurotoxic crises absent in BCAT2 deficiency
38682838	*Psychosis & non-compliance in adult MSUD*	Human clinical (differential)	Identifies leucine as primary neurotoxic metabolite
1404884	*Metabolic disorders of BCAAs due to enzyme mutation*	Review	Context for aminotransferase vs dehydrogenase defects
37494004	*Oral enzyme therapy for MSUD*	Model/NHP	Neighboring therapeutic concept (MSUD, not directly applicable)

How the evidence converges: The human defining paper (P 25653144) establishes genotype, inheritance, functional loss of function, clinical/radiological phenotype,

cofactor responsiveness, and the MSUD exclusion — the entire clinical-genetic core. The two mouse papers (P 17767905) P 26351290 independently confirm the gene identity, pathway position, and biochemical consequence, while revealing that the CNS phenotype is species-dependent. The MSUD literature (P 42329506 P 41257986) P 38682838 provides the essential mechanistic contrast that explains *why* BCAT2 deficiency is milder (upstream block, no BCKA accumulation) and supplies the rationale for dietary management.

Limitations and Knowledge Gaps

1. **Single human case (n=1).** Essentially all human clinical knowledge derives from one molecularly confirmed patient (PMID: 25653144). Phenotype spectrum, penetrance, expressivity, natural history, prevalence, sex ratio, and age distribution cannot be reliably estimated. No additional molecularly confirmed cases have appeared since 2015.
2. **Historical vs modern entity ambiguity.** The classic 1960s "hypervalinemia" and "hyperleucine-isoleucinemia" cases were defined biochemically without molecular confirmation; their precise relationship to *BCAT2* genotype is inferred, not proven.
3. **Model–human phenotype divergence.** The *Bcat2* knockout mouse does not reproduce the human leukoencephalopathy, limiting its utility for studying the CNS pathology and leaving the brain injury mechanism largely inferential.
4. **Mechanism of white-matter injury unresolved.** Whether BCAA elevation per se, glutamate-nitrogen shuttle disturbance, or another route causes the leukoencephalopathy is not established in humans.
5. **Treatment evidence is anecdotal.** Vitamin B6 efficacy and dietary-restriction benefit rest on a single patient and analogy to MSUD, respectively; no trials exist.

6. **Variant classification depth.** Only two variants are known; the full allelic spectrum, genotype-phenotype correlations, and any modifier (e.g., *BCAT1*) effects are unknown.

Proposed Follow-up Experiments / Actions

1. **International case-finding.** Query GeneMatcher, metabolic-disease consortia, and newborn-screening programs for additional biallelic *BCAT2* patients to build a case series and define the phenotype spectrum, penetrance, and natural history.
2. **Re-genotype historical cases.** Where biobanked DNA exists from classic 1960s–90s hypervalinemia/hyperleucine-isoleucinemia patients, perform *BCAT2* sequencing to confirm the nosological unification.
3. **CNS-focused model.** Develop a brain- or oligodendrocyte-targeted *Bcat2* conditional knockout (or patient iPSC-derived oligodendrocyte/organoid model) to interrogate the leukoencephalopathy mechanism that the global mouse KO fails to capture.
4. **Functional variant panel.** Systematically assay BCAT2 enzyme activity and PLP-binding for p.Arg170Gln, p.Glu264Lys, and additional candidate variants to quantify residual activity and predict B6-responsiveness (structure-function; the PLP-binding pocket is a key target).
5. **Prospective treatment documentation.** For any newly identified patient, prospectively record B6 dose-response, dietary BCAA titration, and serial MRI to establish an evidence-based management protocol.
6. **Metabolomic profiling.** Perform plasma/CSF metabolomics in patients to confirm the predicted signature (elevated BCAAs, *absence* of BCKA/alloisoleucine), cementing the biochemical distinction from MSUD and supporting a diagnostic algorithm.

Consensus Answer

Hypervalinemia and Hyperleucine-Isoleucinemia (MONDO:0100058) is an ultra-rare autosomal recessive inborn error of branched-chain amino-acid metabolism caused by biallelic loss-of-function mutations in *BCAT2*, the gene encoding mitochondrial branched-chain aminotransferase (BCATm; chr19q13.33), first molecularly defined by Wang et al. 2015 (PMID: 25653144) via compound-heterozygous p.Arg170Gln and p.Glu264Lys variants. Because BCAT2 catalyzes the first, shared catabolic step of valine, leucine and isoleucine — upstream of the

BCKDH complex defective in maple syrup urine disease — its deficiency elevates all three BCAAs without accumulating neurotoxic branched-chain keto acids, producing a comparatively mild, chronic neurometabolic phenotype (headache, mild memory impairment, symmetric cerebral white-matter abnormalities) that is responsive to vitamin B6 (pyridoxine) cofactor supplementation plus dietary BCAA restriction.
